

● EASY

● ADVANCED MATH

Advanced Math

10 questions · ~15 min

11

Q1

ADVANCED MATH

The function f is defined by $fx = 10x^2 - 32x - 152$. What is the value of $f0$?

- A. -152
- B. -32
- C. 0
- D. 10

Q2

ADVANCED MATH

$$8x^2 - 40 = 32$$

What is the positive solution to the given equation?

- A. 3
- B. 4
- C. 9
- D. 72

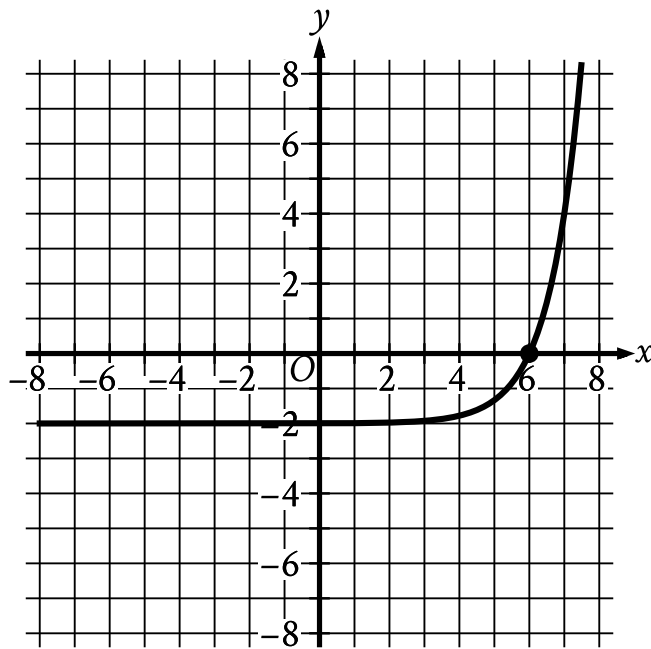
Which of the following expressions is equivalent to $2a^2(a+3)$?

A. $5a^3$

B. $8a^5$

C. $2a^3+3$

D. $2a^3+6a^2$



- Moving from left to right:
 - The curve passes from quadrant 3 to quadrant 4 to quadrant 1.
 - In quadrant 3, the curve trends up gradually to the y axis.
 - In quadrant 4, the curve trends up gradually to the point (6 comma 0).
 - In quadrant 1, the curve trends up sharply.
- As x decreases, the curve approaches the line y equals negative 2.
- The curve passes through the following points:
 - (5 comma negative four thirds)
 - (6 comma 0)

What is the x -coordinate of the x -intercept of the graph shown?

STUDENT-PRODUCED RESPONSE

0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q5

ADVANCED MATH

If $(x+5)^2=4$, which of the following is a possible value of x ?

- A. 1
- B. -1
- C. -2
- D. -3

Q6

ADVANCED MATH

Which of the following is equivalent to $2x^3 + 4$?

A. $4(x^3 + 4)$

B. $4(x^3 + 2)$

C. $2(x^3 + 4)$

D. $2(x^3 + 2)$

Q7

ADVANCED MATH

The function f is defined by $f(x) = 4 + \sqrt{x}$. What is the value of $f(144)$?

A. 0

B. 16

C. 40

D. 76

Q8

ADVANCED MATH

$$b = 42cf$$

The given equation relates the positive numbers b , c , and f . Which equation correctly expresses c in terms of b and f ?

A. $c = \frac{b}{42f}$

B. $c = \frac{b - 42}{f}$

C. $c = 42bf$

D. $c = 42 - b - f$

Q9

ADVANCED MATH

$$(x + 5) + (2x - 3)$$

Which of the following is equivalent to the given expression?

A. $3x - 2$

B. $3x + 2$

C. $3x - 8$

D. $3x + 8$

$$hx = x^2 - 3$$

Which table gives three values of x and their corresponding values of hx for the given function h ?

A.

x	1	2	3
hx	4	5	6

B.

x	1	2	3
hx	-2	1	6

C.

x	1	2	3
hx	-1	1	3

D.

x	1	2	3
hx	-2	1	3